

Harish-Chandra Research Institute

Algebraic Geometry

Assignment 2

M.Sc. Mathematics, Year II •

Unless stated otherwise, $k = \bar{k}$ is an algebraically closed field with $\text{char } k \neq 2, 3$. Here $\Gamma(V)$ is the coordinate ring of a variety V , $\mathcal{O}_P(V)$ its local ring at P with maximal ideal $\mathfrak{m}_P(V)$, $m_P(F)$ is the multiplicity of a plane curve F at P , and $I(P, F \cap G)$ the intersection number of F and G at P . For a polynomial map T and a polynomial F we write $F^T = F \circ T$, and $V^T = T^{-1}(V)$ for a set V .

Problem 1

Let $\mathcal{O}_P(V)$ be the local ring of a variety V at a point P . Establish a natural one-to-one correspondence between the prime ideals of $\mathcal{O}_P(V)$ and the irreducible subvarieties of V passing through P .

Problem 2

Let $T: \mathbb{A}^n \rightarrow \mathbb{A}^n$ be an affine change of coordinates with $T(P) = Q$. Prove that $\tilde{T}: \mathcal{O}_Q(\mathbb{A}^n) \rightarrow \mathcal{O}_P(\mathbb{A}^n)$, $f \mapsto f^T$, is an isomorphism, and that for a subvariety $V \subseteq \mathbb{A}^n$ with $P \in V^T$, the map $\tilde{T}$ induces an isomorphism $\mathcal{O}_Q(V) \rightarrow \mathcal{O}_P(V^T)$.

Problem 3

Let k be any field (not necessarily algebraically closed). The ring of *formal power series* over k is $k[[X]] = \{\sum_{i=0}^{\infty} a_i X^i : a_i \in k\}$, with $\sum a_i X^i + \sum b_i X^i = \sum (a_i + b_i) X^i$ and $(\sum a_i X^i)(\sum b_i X^i) = \sum c_i X^i$, where $c_i = \sum_{j+l=i} a_j b_l$. Prove that $k[[X]]$ is a ring containing $k[X]$ as a subring, and that it is a discrete valuation ring with uniformizing parameter X . Its quotient field is denoted $k((X))$.

Problem 4

Let $V \subseteq \mathbb{A}^n$ be a variety, $I = I(V) \subseteq k[X_1, \dots, X_n]$, and $P \in V$. Let J be an ideal of $k[X_1, \dots, X_n]$ containing I , and let J' be its image in $\Gamma(V)$. Construct a natural homomorphism

$$\varphi: \mathcal{O}_P(\mathbb{A}^n)/J \mathcal{O}_P(\mathbb{A}^n) \longrightarrow \mathcal{O}_P(V)/J' \mathcal{O}_P(V),$$

and prove that φ is an isomorphism. In particular, $\mathcal{O}_P(\mathbb{A}^n)/I \mathcal{O}_P(\mathbb{A}^n) \cong \mathcal{O}_P(V)$.

Problem 5

Let F be a curve of degree n with a point P of multiplicity n . Prove that F is a product of n lines through P (not necessarily distinct).

Problem 6

Let $F \in k[X, Y]$ be a nonzero polynomial and let $T: \mathbb{A}^2 \rightarrow \mathbb{A}^2$ be a polynomial map with $T(Q) = P$, and write $T = (T_1, T_2)$. Let $J_Q T = (\partial T_i / \partial X_j(Q))$ be the Jacobian matrix of T at Q .

- (a) Prove that $m_Q(F^T) \geq m_P(F)$.
 (b) Prove that $m_Q(F^T) = m_P(F)$ whenever $J_Q T$ is invertible.
 (c) Show that the converse of (b) fails: take $T = (X^2, Y)$, $F = Y - X^2$ and $P = Q = (0, 0)$.
 (d) Parts (b) and (c) are local statements. The following shows that invertibility of the Jacobian at *every* point does not force a polynomial map to be an isomorphism of varieties. Let $F = (F_1, F_2, F_3): \mathbb{A}^3 \rightarrow \mathbb{A}^3$ be the polynomial map

$$\begin{aligned} F_1 &= (1 + XY)^3 Z + Y^2(1 + XY)(4 + 3XY), \\ F_2 &= Y + 3X(1 + XY)^2 Z + 3XY^2(4 + 3XY), \\ F_3 &= 2X - 3X^2 Y - X^3 Z. \end{aligned}$$

- (i) Prove that $\det(J_Q F) = -2$ for every $Q \in \mathbb{A}^3$; since $\text{char } k \neq 2$, the matrix $J_Q F$ is invertible at every point.
 (ii) Verify that the three distinct points $(0, 0, -\frac{1}{4})$, $(1, -\frac{3}{2}, \frac{13}{2})$ and $(-1, \frac{3}{2}, \frac{13}{2})$ all have image $(-\frac{1}{4}, 0, 0)$. Deduce that F is not injective, and hence that there is no polynomial map $G: \mathbb{A}^3 \rightarrow \mathbb{A}^3$ with $G \circ F = \text{id}$.

Problem 7

- (a) Let $\mathcal{O} = \mathcal{O}_P(\mathbb{A}^2)$ for a point $P \in \mathbb{A}^2$ and $\mathfrak{m} = \mathfrak{m}_P(\mathbb{A}^2)$. Compute $\chi(n) = \dim_k(\mathcal{O}/\mathfrak{m}^n)$.
 (b) Let $\mathcal{O} = \mathcal{O}_P(\mathbb{A}^r)$. Prove that $\chi(n)$ is a polynomial of degree r in n with leading coefficient $1/r!$.

Problem 8

Let F be a plane curve, P a point of F , and L a line through P . Prove that L is tangent to F at P if and only if $I(P, F \cap L) > m_P(F)$.

Problem 9

Let P be a double point of a curve F , and suppose F has exactly one tangent line L at P .

- (a) Prove that $I(P, F \cap L) \geq 3$. One says that F has an (*ordinary*) *cusp* at P if $I(P, F \cap L) = 3$.
 (b) Suppose $P = (0, 0)$ and $L = Y$. Prove that P is a cusp if and only if $F_{XXX}(P) \neq 0$.

Problem 10

Let $F \in k[X_1, \dots, X_{n+1}]$; write $F = \sum_i F_i$ with F_i a form of degree i . Let $P \in \mathbb{P}^n(k)$ and suppose $F(x_1, \dots, x_{n+1}) = 0$ for every choice of homogeneous coordinates $(x_1, \dots, x_{n+1})$ of P . Prove that $F_i(x_1, \dots, x_{n+1}) = 0$ for each i and for every choice of homogeneous coordinates of P .